

FRACTIONAL TRIANGLE HYPERSURFACES: AUTOMATIC COX MODELS AND DIVISOR CLASSIFICATION

KENJI HASHIMOTO, HWAYOUNG LEE, AND KAZUSHI UEDA

ABSTRACT. Let

$$T(p, q, r) = \mathbb{C}[X, Y, Z]/(X^p + Y^q + Z^r)$$

carry its Geigle–Lenzing grading, let $\vec{\omega}$ be the dualizing element, and suppose that $\delta\vec{\tau} = \vec{\omega}$. We classify, for every positive index δ , the root section rings $R(p, q, r; \vec{\tau}) = \bigoplus_{m \geq 0} T_{m\vec{\tau}}$ that are hypersurfaces. We begin with a self-contained Geigle–Lenzing-lattice proof of the known three-point root criterion, a finite diagonal-invariant description, and the identities

$$h - a - b - c = \delta, \quad \frac{h}{abc} = \frac{1}{\delta} \left(1 - \frac{1}{p} - \frac{1}{q} - \frac{1}{r} \right)$$

for any hypersurface presentation of weights $(a, b, c; h)$. The main structural result is that every such root hypersurface is automatically Cox-invertible: after a graded change of generators its relation is one of the five ternary Fermat–chain–loop types, has no residual diagonal symmetry, and admits an explicit monomial lift to $X^p + Y^q + Z^r$. We then turn the five atomic equations into an exact finite divisor parametrization, including closed coprimality tests. This gives an unconditional classification at arbitrary index; in particular, a fractional triangle hypersurface exists whenever $6 \nmid \delta$. We record the resulting census through index thirty and provide deterministic certificates and verification code for the enumeration.

1. INTRODUCTION

Let $p, q, r \geq 2$ and assume

$$(1.1) \quad \nu = 1 - \frac{1}{p} - \frac{1}{q} - \frac{1}{r} > 0.$$

The Brieskorn–Pham ring

$$(1.2) \quad T = T(p, q, r) = \mathbb{C}[X, Y, Z]/(X^p + Y^q + Z^r)$$

is graded by

$$(1.3) \quad L = L(p, q, r) = \frac{\mathbb{Z}\vec{x} \oplus \mathbb{Z}\vec{y} \oplus \mathbb{Z}\vec{z} \oplus \mathbb{Z}\vec{c}}{(p\vec{x} - \vec{c}, q\vec{y} - \vec{c}, r\vec{z} - \vec{c})}, \quad \deg_L(X, Y, Z) = (\vec{x}, \vec{y}, \vec{z}).$$

The associated Geigle–Lenzing curve $\mathbb{X}(p, q, r)$ is a smooth stacky $\mathbb{P}^1$ with stabilizer orders p, q, r [10]. Its dualizing element is

$$(1.4) \quad \vec{\omega} = \vec{c} - \vec{x} - \vec{y} - \vec{z}.$$

We study a positive root

$$(1.5) \quad \delta\vec{\tau} = \vec{\omega}$$

and its section ring

$$(1.6) \quad R(p, q, r; \vec{\tau}) = \bigoplus_{m \geq 0} T_{m\vec{\tau}}.$$

Date: Version 04, 15 August 2026.

2020 Mathematics Subject Classification. 14H45, 14J17, 13A02, 32S25.

Key words and phrases. weighted projective line, triangle singularity, canonical root, Cox ring, invertible polynomial, divisor parametrization.

We call $(p, q, r; \delta, \vec{\tau})$ a *fractional triangle datum*. If its root section ring has a primitive minimal three-generator hypersurface presentation, its graded-isomorphism class is called a *fractional triangle hypersurface*. Signatures are taken up to permutation. Atomic presentations introduced later are witnesses, not additional classes: the class census identifies all witnesses that give the same graded ring. The separate witness certificate uses a supplementary normal-form convention in which an L3 witness retains a choice of cyclic starting vertex, while FFF permutations and the L2F swap are already quotiented. The adjective *marked* refers only to this certificate convention and does not define an additional invariant.

For $\delta = 1$, (1.6) is the canonical triangle ring. Its hypersurface cases are the fourteen exceptional unimodal triangle singularities in Dolgachev’s list; see [4, 24] and the two-dimensional case of [11]. Allowing $\delta > 1$ replaces the canonical polarization by an integral δ -th root of it.

These rings also belong to the general theory of section rings of rational divisors on $\mathbb{P}^1$. Via the coarse moduli map there is a rational divisor $D_{\vec{\tau}}$ on $\mathbb{P}^1$, determined by $\mathcal{O}_{\mathbb{X}}(\vec{\tau})$, for which

$$R(p, q, r; \vec{\tau}) \cong \bigoplus_{m \geq 0} H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(\lfloor m D_{\vec{\tau}} \rfloor)).$$

O’Dorney studies this ambient class of rings [18, Theorems 4, 6, 8, 14, and 17]: he gives complete presentations for support of size at most two, general bounds on generator and relation degrees, and stability results for generators and Gröbner leading terms. Those results do not give the present root-constrained three-point hypersurface classification; the canonical-root condition and the three-generator hypothesis are what make an exact arbitrary-index classification possible here.

The spin case, which is index $\delta = 2$ in our notation, has a sharper overlap with earlier work. Landesman–Ruhm–Zhang [14, Table 7.5] list the exceptional no-log genus-zero spin rings. Their six hyperbolic three-point entries have spin signature and generator/relation degrees

$$\begin{array}{ll} (0; 3, 3, 5; 0) \mapsto (3, 10, 15; 30) & (0; 3, 3, 7; 0) \mapsto (3, 7, 12; 24) \\ (0; 3, 3, 9; 0) \mapsto (3, 7, 9; 21) & (0; 3, 5, 5; 0) \mapsto (3, 5, 10; 20) \\ (0; 3, 5, 7; 0) \mapsto (3, 5, 7; 17) & (0; 5, 5, 5; 0) \mapsto (3, 5, 5; 15). \end{array}$$

These are exactly the six systems in our index-two census. Thus no index-two weight system is claimed to be new. The contribution here is the automatic converse at arbitrary index, the Cox lifts in the classified setting, and the exact divisor parametrization.

The existence of such roots is already understood in the Fuchsian-lift language. For signature $(g; p_1, \dots, p_s)$, Natanzon–Pratoussevitch [16, Proposition 4.10] prove that lifts to the m -fold connected cover exist precisely when

$$\gcd(m, p_i) = 1 \quad (1 \leq i \leq s), \quad (p_1 \cdots p_s) \left(\sum_{i=1}^s \frac{1}{p_i} - (2g - 2) - s \right) \equiv 0 \pmod{m},$$

and that there are m^{2g} such lifts. Dualizing their tangent-root convention gives the canonical-root convention used here. For $(g, s, m) = (0, 3, \delta)$ their congruence is $\delta \mid pqr\nu$. Under $\gcd(\delta, N) = 1$, the integer pqr/N is invertible modulo δ , so this is equivalent to $\delta \mid N\nu$; their count is one. Thus Theorem 1.1 is a self-contained algebraic specialization and reproof of that criterion in the Geigle–Lenzing lattice, included in the form needed for the invariant-ring calculation and the maximal root. The automorphy-factor/root correspondence goes back to Dolgachev [5].

Put $N = \text{lcm}(p, q, r)$, $k = N\nu$, and let

$$(1.7) \quad d_{\max} = \max\{e > 0 : e \mid k, \gcd(e, N) = 1\}.$$

For reference we isolate the elementary grading constraints that drive the classification.

Theorem 1.1 (Root criterion). *There is a unique $\vec{\omega} \in L$ such that $\vec{\omega} = d_{\max} \vec{\omega}$. For a positive integer δ , a root $\delta \vec{\tau} = \vec{\omega}$ exists if and only if*

$$\delta \mid k \quad \text{and} \quad \gcd(\delta, N) = 1,$$

equivalently if and only if $\delta \mid d_{\max}$. The root is unique and equals $\vec{\tau} = (d_{\max}/\delta) \vec{\omega}$.

Theorem 1.2 (Numerical identities). *If the root ring has a primitive minimal hypersurface presentation*

$$R(p, q, r; \vec{\tau}) \cong \mathbb{C}[x, y, z]/(f), \quad \deg(x, y, z; f) = (a, b, c; h),$$

then

$$(1.8) \quad h - a - b - c = \delta, \quad \frac{h}{abc} = \frac{\nu}{\delta}.$$

The identities in Theorem 1.2 are the specialization to the present three-point root data of the standard Gorenstein–DPD relations $a(R) = h - a - b - c$ and $\deg D = h/(abc)$; compare [25, Proposition 1.1]. We give an invariant-theoretic proof in Section 3 in order to fix conventions and to connect the formula directly with the root index.

The central point is that no hypothesis on the number or shape of the monomials in f is needed.

Theorem 1.3 (Automatic Cox presentation). *Every three-variable fractional triangle hypersurface is Cox-invertible. Equivalently, one can choose a primitive weighted atomic presentation of one of the five types FFF, C2F, L2F, C3, and L3 whose exponent determinant has absolute value h , and which has an explicit monomial lift to $X^p + Y^q + Z^r$.*

The proof uses the polarized stacky Proj to recover $(\mathbb{X}(p, q, r), \mathcal{O}(\vec{\tau}))$ from the abstract graded ring. A deformation-rigidity argument reduces the defining equation to the ternary graph classification. The two branched noninvertible graphs have either the wrong number of stacky points or positive coarse genus; the few atomic models with a nonprimitive exponent determinant admit explicit graded mutations. This produces an atomic presentation with no residual diagonal symmetry, from which the Cox lift is recovered.

The passage from a regular weight system to fractional-canonical orbifold geometry is classical. For $W = (a, b, c; h)$, put $\varepsilon_W = a + b + c - h$; for the weights considered here, $\varepsilon_W = -\delta$. Saito associates to a regular system a generic isolated weighted-homogeneous hypersurface, its quotient orbifold curve, an arithmetically determined genus and signature, and a fractional power of the canonical bundle; for the genus and signature see [20, Theorem 6], and for the fractional-canonical bundle see [21, Section 11, especially Facts 5–6]. Saito’s convention is dual to the positive section-ring polarization used here. Thus neither the fractional-canonical interpretation nor the recovery of orbifold data from a regular weight system is itself new. The distinction needed here is that we start from an arbitrary relation in the three-point root subclass and force it into a primitive principal-lattice model with trivial residual symmetry.

The five atomic types and their explicit principal-lattice Cox formulas are also known. They occur in the same theory and in the work of Ebeling–Takahashi; see [20] and [8, Propositions 1 and 4 and Table 3] for the five types and monomial embeddings. In the canonical/reduced-weight notation of [9, Sections 1–2], the canonical degree is $d = |\det E|$, division by the common factor c_f gives the reduced degree $h = d/c_f$. Writing G_f^{fin} for the finite diagonal symmetry group and $G_0 = \langle g_0 \rangle$ for its exponential-grading subgroup, they identify $G_0^T \simeq \text{Hom}(G_f^{\text{fin}}/G_0, \mathbb{C}^*)$ and prove $|G_0^T| = |G_f^{\text{fin}}/G_0| = c_f$ [9, Propositions 3 and 5]. Thus $|\det E| = h$ is precisely the known case $c_f = 1$. More directly, Takahashi

[22, Main Theorem 5.1 and Appendix, Section 8] constructs the principal-lattice three-generator ring and explicit Cox monomials for the same five types under the dual-type hypothesis. The canonical case $\delta = 1$ recovers the fourteen hypersurfaces in [11].

The abstract and introduction of [22] also announce a joint extension with H. Kajiura and K. Saito from regular systems of dual type to all genus-zero regular systems; its reference KST3 describes that project as work in progress. As of August 2026, we are not aware of a published or publicly accessible completion of that announcement. The announcement itself does not formulate an automatic mutation or Cox normal-form theorem. Since the promised paper does not appear to be publicly accessible, the full extent of its overlap with the present results cannot be assessed from the announcement alone. We record it because it may overlap with the stack-identification part of the argument, but we do not rely on or make any assertion about unpublished material.

The new content of Theorem 1.3 is the converse at arbitrary index: an *arbitrary* fractional triangle hypersurface, with no atomic or invertibility hypothesis on its initial relation, is forced after graded mutation into one of these principal-lattice models with trivial residual symmetry. Together with the exact divisor arithmetic, this promotes the known constructions to an arbitrary-index classification.

Combining this theorem with elementary factorizations of the five atomic degree equations gives our classification theorem.

Theorem 1.4 (Arbitrary-index classification). *For every $\delta > 0$, all fractional triangle hypersurfaces of index δ are given by the five finite divisor constructions in Theorem 5.6, subject only to the closed coprimality conditions in Proposition 5.7 and identification of repeated graded presentations.*

Thus the result is a structural classification, rather than a bounded computer search. It immediately gives explicit models at every index not divisible by six. As a small numerical check, exact divisor enumeration recovers the classical 14 types at $\delta = 1$ and gives 506 graded isomorphism classes for $2 \leq \delta \leq 30$.

Remark 1.5 (Comparison with Watanabe’s arXiv-v1 list). We compared Section 3 of the versioned preprint [26], namely `arXiv:1401.0789v1`, with the fractional-triangle subclass classified here. This is a comparison with the arXiv version submitted on 4 January 2014, not with an independently checked copy of the published journal article. Using sorted primitive weights and reading the punctuation $(4, 10.17; 34)$ as $(4, 10, 17; 34)$, the numbers “arXiv-v1/current classification” for $a(R) = 1, \dots, 6$ are

$$14/14, \quad 6/6, \quad 7/8, \quad 7/8, \quad 17/21, \quad 0/0.$$

Within this stated subclass, range, and version, the exact difference is

$$a(R) = 3 : (4, 10, 13; 30),$$

$$a(R) = 4 : (5, 6, 9; 24),$$

$$a(R) = 5 : (6, 8, 19; 38), (4, 7, 16; 32), (4, 7, 12; 28), (4, 7, 9; 25).$$

They are realized explicitly in Remark 6.3. This does not affect Watanabe’s finiteness theorem [26, Theorem 2.1], and makes no assertion outside the stated subclass or about a later correction. Four systems occur explicitly in earlier tables: [6, Table 2(a)] contains $(5, 6, 9; 24)$ and $(6, 8, 19; 38)$; in the K3 notation of [7, Table 3], the rows $(3; 4, 10, 13; 30)$ and $(5; 4, 7, 9; 25)$ give the first and sixth systems after suppressing the initial K3 weight. No underlying singularity is claimed to be newly discovered. The comparison script, extracted keys, and six-row difference certificate are included in the supplementary archive.

Section 2 proves the root criterion, and Section 3 proves the invariant-ring description and the numerical identities. Section 4 proves automatic Cox-invertibility after recovering

the polarized triangle from its section ring. Section 5 gives the five atomic models and the divisor classification. Section 6 records concise existence and low-index consequences.

2. THE GRADING GROUP AND ROOTS OF THE DUALIZING ELEMENT

For uniform notation in the calculations below, set $n = 3$, $(p_1, p_2, p_3) = (p, q, r)$, and $(\vec{x}_1, \vec{x}_2, \vec{x}_3) = (\vec{x}, \vec{y}, \vec{z})$.

We use the degree homomorphism

$$(2.1) \quad \deg : L \longrightarrow \mathbb{Q}, \quad \deg \vec{c} = 1, \quad \deg \vec{x}_i = \frac{1}{p_i}.$$

Its image is $\frac{1}{N}\mathbb{Z}$. For every $\vec{v} \in L$ one has

$$(2.2) \quad N\vec{v} = N \deg(\vec{v})\vec{c}.$$

It follows that $\ker(\deg) = \text{tors}(L)$ and that its exponent divides N . Thus

$$(2.3) \quad 0 \longrightarrow \text{tors}(L) \longrightarrow L \xrightarrow{\deg} \frac{1}{N}\mathbb{Z} \longrightarrow 0$$

is exact. Notice also that

$$(2.4) \quad L/\mathbb{Z}\vec{c} \cong \bigoplus_{i=1}^n \mathbb{Z}/p_i\mathbb{Z}, \quad \vec{\omega} \equiv -\sum_i \vec{x}_i \pmod{\mathbb{Z}\vec{c}}.$$

Lemma 2.1. *If e is coprime to N , multiplication by e on L is injective.*

Proof. An element killed by e has degree zero, hence belongs to $\text{tors}(L)$. It is also killed by N by (2.2). Since $\gcd(e, N) = 1$, it is zero. $\square$

Proof of Theorem 1.1. Suppose first that $\delta\vec{\tau} = \vec{\omega}$. Reducing modulo $\mathbb{Z}\vec{c}$ in (2.4) shows that multiplication by δ must be invertible in every $\mathbb{Z}/p_i\mathbb{Z}$. Hence

$$(2.5) \quad \gcd(\delta, N) = 1.$$

Taking degrees and multiplying by N gives

$$k = N \deg \vec{\omega} = \delta N \deg \vec{\tau},$$

so $\delta \mid k$. Therefore $\delta \mid d_{\max}$.

It remains to construct the maximal root. Choose integers b_i satisfying $b_i p_i \equiv 1 \pmod{d_{\max}}$. Since $d_{\max} \mid k$ and $\gcd(d_{\max}, N) = 1$,

$$N \left(1 - \sum_i b_i \right) \equiv N - \sum_i \frac{N}{p_i} = k \equiv 0 \pmod{d_{\max}},$$

and hence $1 - \sum_i b_i$ is divisible by $d_{\max}$. The element

$$(2.6) \quad \vec{\omega} = \frac{1 - \sum_i b_i}{d_{\max}} \vec{c} + \sum_i \frac{b_i p_i - 1}{d_{\max}} \vec{x}_i$$

belongs to L and satisfies $d_{\max} \vec{\omega} = \vec{\omega}$.

If $\delta \mid d_{\max}$, then $\vec{\tau} = (d_{\max}/\delta) \vec{\omega}$ is a δ -th root. Conversely, any root has $\delta \mid d_{\max}$ by the first paragraph, and Lemma 2.1 applied to

$$\delta \left(\vec{\tau} - \frac{d_{\max}}{\delta} \vec{\omega} \right) = 0$$

proves uniqueness. $\square$

Remark 2.2. Write a root in normal form modulo $\mathbb{Z}\vec{c}$ as $\vec{\tau} \equiv \sum_i s_i \vec{x}_i$. Equation (1.5) gives

$$(2.7) \quad \delta s_i \equiv -1 \pmod{p_i}.$$

In particular the character of $\vec{\tau}$ at the i -th stacky point is faithful.

3. INVARIANT RINGS, THE CANONICAL SHIFT, AND MULTIPLICITY

Fix a root $\delta\vec{\tau} = \vec{\omega}$ and put

$$(3.1) \quad \mu = \deg \vec{\tau} = \frac{\nu}{\delta}, \quad u = N\mu = \frac{k}{\delta} \in \mathbb{Z}_{>0}.$$

The quotient $L/\mathbb{Z}\vec{\tau}$ is finite. Its Cartier dual

$$(3.2) \quad G_{\vec{\tau}} := \text{Hom}(L/\mathbb{Z}\vec{\tau}, \mathbb{C}^*)$$

acts diagonally on T . We emphasize that $L/\mathbb{Z}\vec{\tau}$ is the character group of $G_{\vec{\tau}}$, not canonically the group $G_{\vec{\tau}}$ itself.

Proposition 3.1. *The group and its order are*

$$(3.3) \quad G_{\vec{\tau}} = \left\{ \text{diag}(\alpha_1, \dots, \alpha_n) : \alpha_i^{p_i} = \rho := \prod_{j=1}^n \alpha_j, \quad \rho^u = 1 \right\},$$

$$(3.4) \quad |G_{\vec{\tau}}| = \mu \prod_{i=1}^n p_i.$$

Moreover,

$$(3.5) \quad R(\mathbf{p}, \vec{\tau}) = T^{G_{\vec{\tau}}}.$$

Proof. A character annihilating $\vec{\tau}$ annihilates $\vec{\omega} = \delta\vec{\tau}$. The relations in L and the identity $\alpha(\vec{\omega}) = 1$ give $\alpha_i^{p_i} = \alpha(\vec{\omega}) = \prod_j \alpha_j = \rho$. Since $N\vec{\tau} = u\vec{\omega}$, one also has $\rho^u = 1$.

Conversely, the displayed equations imply $\alpha(\vec{\omega}) = 1$ and $\alpha(N\vec{\tau}) = 1$. Thus $\alpha(\vec{\tau})^\delta = \alpha(\vec{\tau})^N = 1$; by (2.5), $\alpha(\vec{\tau}) = 1$. This proves (3.3).

The torsion group in (2.3) has order $\prod_i p_i / N$, while the image of $\vec{\tau}$ has index u in $\frac{1}{N}\mathbb{Z}$. Therefore

$$|L/\mathbb{Z}\vec{\tau}| = u \frac{\prod_i p_i}{N} = \mu \prod_i p_i,$$

which proves (3.4). Finally, a homogeneous element of T is fixed by all characters of $L/\mathbb{Z}\vec{\tau}$ precisely when its L -degree belongs to $\mathbb{Z}\vec{\tau}$. Its nonnegative-degree part is (3.5). $\square$

The Brieskorn–Pham hypersurface T is normal: its only singular point is the origin and $\dim T \geq 2$. Since finite-group invariants preserve normality, $R(\mathbf{p}, \vec{\tau})$ is finitely generated and normal. Give T the rational $\vec{\tau}$ -degree

$$(3.6) \quad \deg_{\vec{\tau}} X_i = \frac{1}{\mu p_i}, \quad \deg_{\vec{\tau}} \left(\sum_i X_i^{p_i} \right) = \frac{1}{\mu}.$$

The invariant part has integral degrees. Molien’s formula gives the following expression. Indeed, if $S = \mathbb{C}[X_1, \dots, X_n]$ and $F = \sum_i X_i^{p_i}$, then $g(F) = \rho(g)F$; the equivariant exact sequence

$$0 \longrightarrow S(-1/\mu) \otimes \mathbb{C}_\rho \xrightarrow{\cdot F} S \longrightarrow T \longrightarrow 0$$

accounts for the semi-invariant numerator:

$$(3.7) \quad P_R(t) = \frac{1}{|G_{\vec{\tau}}|} \sum_{g \in G_{\vec{\tau}}} \frac{1 - \rho(g)t^{1/\mu}}{\prod_{i=1}^n (1 - \alpha_i(g)t^{1/(\mu p_i)}}.$$

Proposition 3.2. *If $R = R(\mathbf{p}, \vec{\tau})$ has a hypersurface presentation*

$$R \cong \mathbb{C}[z_1, \dots, z_n]/(f), \quad \deg z_i = a_i, \quad \deg f = h,$$

then

$$(3.8) \quad h - \sum_i a_i = \delta, \quad \frac{h}{\prod_i a_i} = \mu^{n-2}.$$

With the convention $M(s)_m = M_{m+s}$, the hypersurface canonical-module formula also gives $K_R \cong R(\delta)$.

Proof. Replace t by t^{-1} in (3.7), factor the powers of t , and reindex the sum by $g \mapsto g^{-1}$. Since

$$\frac{1}{\mu} - \sum_i \frac{1}{\mu p_i} = \frac{\nu}{\mu} = \delta, \quad \rho(g) = \prod_i \alpha_i(g),$$

we obtain

$$(3.9) \quad P_R(t^{-1}) = (-1)^{n-1} t^{-\delta} P_R(t).$$

For the hypersurface Hilbert series

$$P_R(t) = \frac{1 - t^h}{\prod_i (1 - t^{a_i})},$$

the same calculation gives the exponent $-(h - \sum_i a_i)$. This proves the first identity.

For the second identity, consider the pole at $t = 1$. A nonidentity element of $G_{\vec{\tau}}$ cannot fix $n - 1$ coordinates: if all but α_i are 1, then $\rho = 1$ from any fixed coordinate and hence $\alpha_i = 1$ as well. Thus only the identity contributes to the pole of order $n - 1$. Its contribution before division by $|G_{\vec{\tau}}|$ has leading coefficient

$$\frac{1/\mu}{\prod_i (1/(\mu p_i))} = \mu^{n-1} \prod_i p_i.$$

Using (3.4), the leading coefficient of $P_R(t)$ is μ^{n-2} . The hypersurface expression has leading coefficient $h/\prod_i a_i$, and the result follows. $\square$

4. RECOVERY AND AUTOMATIC COX-INVERTIBILITY

Definition 4.1. A three-variable fractional triangle ring is *Cox-invertible* if it has a primitive minimal graded presentation

$$R(\mathbf{p}, \vec{\tau}) = \mathbb{C}[x, y, z]/(f), \quad \deg(x, y, z; f) = (a, b, c; h),$$

with the following properties.

- (1) The generators x, y, z are images of monomials M_1, M_2, M_3 in X, Y, Z .
- (2) The relation f is a sum of three monomials, its exponent matrix A is nonsingular, and $|\det A| = h$.
- (3) For a monomial M_0 ,

$$(4.1) \quad f(M_1, M_2, M_3) = M_0(X^p + Y^q + Z^r).$$

Here primitive means $\gcd(a, b, c) = 1$, and minimal means that x, y, z form a basis of the irrelevant ideal modulo its square. We identify entries up to a permutation of the signature and a graded isomorphism of presentations.

Lemma 4.2 (Recovery from the root section ring). *Let $\mathbb{X} = \mathbb{X}(p, q, r)$ and $\mathcal{L} = \mathcal{O}_{\mathbb{X}}(\vec{\tau})$, where $\delta\vec{\tau} = \vec{\omega}$ and $\deg \vec{\tau} > 0$. The natural section-ring morphism is a polarized isomorphism*

$$(\mathbb{X}, \mathcal{L}) \xrightarrow{\sim} \left(\text{Proj}_{\text{st}} \bigoplus_{m \geq 0} H^0(\mathbb{X}, \mathcal{L}^m), \mathcal{O}(1) \right) = (\text{Proj}_{\text{st}} R(p, q, r; \vec{\tau}), \mathcal{O}(1)),$$

where $\text{Proj}_{\text{st}} R = [(\text{Spec } R \setminus V(R_+))/\mathbb{G}_m]$.

Proof. Put $S = \bigoplus_{m \geq 0} H^0(\mathbb{X}, \mathcal{L}^m)$ and $\mathcal{Y} = \text{Proj}_{\text{st}} S$. We first recover the stabilizers directly from S . For $P \in \mathbb{X}$, set

$$\Sigma_P = \{m \geq 1 : \text{some } s \in H^0(\mathbb{X}, \mathcal{L}^m) \text{ satisfies } s(P) \neq 0\}.$$

The standard Geigle–Lenzing Cox construction [10] identifies $T_{\vec{v}}$ with $H^0(\mathbb{X}, \mathcal{O}_{\mathbb{X}}(\vec{v}))$. Equivalently, a section is a homogeneous function on the punctured affine Cox cone, and it extends across the vertex because T is a normal surface and the vertex has codimension two. Hence $S = \bigoplus_{m \geq 0} T_{m\vec{\tau}} = R(p, q, r; \vec{\tau})$, as used in the statement.

The stabilizer at the image of P in $\mathcal{Y}$ is

$$(4.2) \quad \{\lambda \in \mathbb{G}_m : \lambda^m = 1 \text{ for every } m \in \Sigma_P\} = \mu_{\gcd(\Sigma_P)}.$$

Indeed, this is the stabilizer in the quotient presentation, computed on a homogeneous affine chart on which a section nonzero at P has been inverted.

Suppose that $P = P_i$ has stabilizer μ_{p_i} , and let $s_i \in \mathbb{Z}/p_i\mathbb{Z}$ be the character of $\mathcal{L}$ on the residual gerbe $\mathcal{G}_i \simeq B\mu_{p_i}$. The root equation gives $\delta s_i \equiv -1 \pmod{p_i}$, so s_i is a unit. A section nonzero at P_i can therefore exist only when $p_i \mid m$. Conversely, let $\mathcal{I}_i$ be the ideal of $\mathcal{G}_i$. For $p_i \mid m$, the target of the evaluation map in

$$0 \longrightarrow \mathcal{I}_i \mathcal{L}^m \longrightarrow \mathcal{L}^m \longrightarrow \mathcal{L}^m|_{\mathcal{G}_i} \longrightarrow 0$$

has a one-dimensional invariant space. A sufficiently divisible power of $\mathcal{L}$ descends to an ample line bundle on the coarse $\mathbb{P}^1$. Serre vanishing, applied in the finitely many residue classes of m , gives $H^1(\mathbb{X}, \mathcal{I}_i \mathcal{L}^m) = 0$ for every sufficiently large multiple of p_i . Thus all such multiples belong to Σ_{P_i} and

$$(4.3) \quad \gcd(\Sigma_{P_i}) = p_i.$$

At a nonstacky point the same evaluation argument gives $\gcd(\Sigma_P) = 1$. Equations (4.2)–(4.3) recover exactly the three stabilizers and the faithful character of $\mathcal{O}(1)$.

Choose M divisible by p, q, r . The line bundle $\mathcal{L}^M$ descends to an ample line bundle on the coarse $\mathbb{P}^1$, so ordinary Proj reconstruction for this Veronese section ring shows that the coarse map underlying $\mathbb{X} \rightarrow \mathcal{Y}$ is an isomorphism. The morphism is representable because its map on every stabilizer is injective, as detected by the faithful pulled-back $\mathcal{O}(1)$ character. It is proper and quasifinite, hence finite, and is birational because the coarse map is an isomorphism and both generic stabilizers are trivial.

Finally, $S = R(p, q, r; \vec{\tau})$ is normal, so $\mathcal{Y}$ is a normal Deligne–Mumford curve. A finite representable birational morphism to a normal Deligne–Mumford stack is an isomorphism: after an étale atlas this is the usual normalization argument. Hence $\mathbb{X} \rightarrow \mathcal{Y}$ is an isomorphism, and its construction identifies $\mathcal{L}$ with $\mathcal{O}(1)$. $\square$

Lemma 4.3 (Stabilizer strata in a tame family of curves). *Let $\pi: \mathcal{C} \rightarrow S$ be a smooth proper tame Deligne–Mumford stack of relative dimension one over a connected complex scheme. Assume that every geometric fiber is connected and that at least one geometric fiber has trivial generic stabilizer. Then every geometric fiber has trivial generic stabilizer and, after an étale cover of S , the locus of nontrivial stabilizers is a disjoint union of gerbes*

$$B\mu_{r_1}, \dots, B\mu_{r_e}$$

over disjoint sections of the coarse curve. In particular, e , the orders r_j , the characters on the relative tangent lines, and the characters of any line bundle on $\mathcal{C}$ are locally constant, up to permutation.

Proof. The relative inertia $I_{\mathcal{C}/S} \rightarrow \mathcal{C}$ is finite and unramified. Its identity section is open because inertia is unramified and closed because $\mathcal{C}$ is separated; hence the complement $I_{\mathcal{C}/S}^*$ is proper over S . Tame stacks are étale-locally quotients by finite linearly reductive

groups [2, Theorem 3.2]. On a quotient chart $[V/G]$, with $V \rightarrow S$ smooth and G finite, the branches of inertia are quotients of fixed loci V^g .

For completeness, $V^g \rightarrow S$ is smooth. Across a square-zero extension with ideal J , the obstruction to fixing a chosen lift lies in $H^1(\langle g \rangle, T_{V/S} \otimes J) = 0$ by linear reductivity over $\mathbb{C}$. Changing the lift by a tangent vector and applying the infinitesimal lifting criterion proves that $V^g \rightarrow S$, and hence $I_{\mathcal{C}/S}^* \rightarrow S$, is smooth. The union of its relative one-dimensional components is open and closed. Its image in S is open by smoothness and closed by properness; if nonempty, it is therefore all of the connected base. It would then meet the fiber assumed to have trivial generic stabilizer, which has no such component. Consequently all fibers of $I_{\mathcal{C}/S}^* \rightarrow S$ are zero-dimensional, which also proves that every fiber of $\mathcal{C}$ has trivial generic stabilizer. Thus $I_{\mathcal{C}/S}^* \rightarrow S$ is proper and smooth of relative dimension zero. Let Z be its relative coarse space. By tameness and compatibility of coarse-space formation with base change [2, Corollary 3.3], the morphism $Z \rightarrow S$ is proper and etale, hence finite etale.

At a stacky point the stabilizer acts faithfully on the relative tangent line: a finite-order automorphism of a formal disc over $\mathbb{C}$ whose derivative is one is the identity, and a kernel would give a nontrivial generic stabilizer in that fiber. The stabilizer is consequently cyclic. The standard local description of a smooth tame orbifold curve (compare [3, Section 2]) gives, after strict henselization and formal completion,

$$[\mathrm{Spec} B[[z]]/\mu_r], \quad \zeta \cdot z = \zeta z,$$

with μ_r acting trivially on B . The coarse curve is locally $\mathrm{Spec} B[[u]]$, where $u = z^r$, and the coarse space of each nonidentity inertia branch is $\mathrm{Spec} B$, mapping by $u = 0$.

Let C be the coarse curve of $\mathcal{C}$ and let $\rho: Z \rightarrow C$ be the induced map. The preceding local model shows that

$$Z \times_C Z \subset Z \times_S Z$$

is, after strict henselization, a union of connected components. Since $Z \times_S Z$ is finite etale over S , this decomposition descends after an etale shrinking; hence the relation of having the same image in C is open and closed. After a further etale base change, its equivalence classes give disjoint stacky sections. The inertia group law identifies the class over a stabilizer μ_r with its $r - 1$ nonidentity elements, and the local model identifies the corresponding stacky locus with the gerbe $B\mu_r$. A character on a line bundle is an element of the discrete group $\mathrm{Hom}(\mu_r, \mathbb{G}_m) = \mathbb{Z}/r\mathbb{Z}$, so it is locally constant; the same applies to the tangent character. $\square$

Lemma 4.4 (Deformation rigidity of a polarized weighted triangle). *Fix primitive positive weights (A, B, C) and a positive degree h , and let U be the locus of quasismooth degree- h forms in $H^0(\mathbb{P}(A, B, C), \mathcal{O}(h))$. If one fiber over U has coarse genus zero and exactly three nontrivial stacky points, then every fiber does. Moreover, any two fibers are isomorphic as polarized stacks, with polarization $\mathcal{O}(1)$.*

Proof. The quasismooth locus U is a nonempty Zariski open subset of a vector space, hence is irreducible. The universal quotient hypersurface

$$\pi: \mathcal{X} = [(V(\mathcal{F}) \setminus \{0\})/\mathbb{G}_m] \longrightarrow U$$

is a smooth proper tame Deligne–Mumford curve. The number of geometric connected components is locally constant in a smooth proper family. It is one on the triangle fiber in the hypothesis, hence one on the connected base U . That fiber also has trivial generic stabilizer, so Lemma 4.3 applies to the universal family and shows that the number of stacky points, their orders, and their tangent and $\mathcal{O}(1)$ characters are constant on U , up to permutation. The coarse genus is constant as well. Indeed, exactness of coarse pushforward for tame stacks and compatibility with base change [2, Theorem 3.2 and

Corollary 3.3] identify the cohomology of the structure sheaf on a fiber with that of its coarse fiber. In a smooth proper connected family $\chi(\mathcal{O})$ is locally constant and $h^0(\mathcal{O}) = 1$, so $h^1(\mathcal{O})$, hence the coarse genus, is locally constant.

Thus, if one fiber is a triangle, every fiber is a smooth orbifold curve with coarse curve $\mathbb{P}^1$ and the same three stabilizer orders. Such a curve, having trivial generic stabilizer, is the iterated root stack of $\mathbb{P}^1$ at its three stacky points; compare [3, Section 2]. Three points on $\mathbb{P}^1$ have no cross-ratio modulus. Finally, the Picard sequence for this root stack is

$$0 \longrightarrow \mathrm{Pic}(\mathbb{P}^1) \longrightarrow \mathrm{Pic}(\mathcal{X}_u) \longrightarrow \bigoplus_{i=1}^3 \mathbb{Z}/r_i\mathbb{Z} \longrightarrow 0.$$

The image in the last group records the three stabilizer characters, and the orbifold degree then determines the remaining coarse degree because $\mathrm{Pic}^0(\mathbb{P}^1) = 0$. The characters are constant as above, while $\deg \mathcal{O}(1) = h/(ABC)$ is fixed. An automorphism of $\mathbb{P}^1$ matching the three points therefore lifts to an isomorphism of the root stacks carrying one $\mathcal{O}(1)$ to the other. $\square$

Lemma 4.5 (Stacky points and adjunction). *Let*

$$\mathcal{C} = [(V(F) \setminus \{0\})/\mathbb{G}_m] \subset \mathbb{P}(A, B, C)$$

be a quasismooth curve for primitive weights (A, B, C) and degree h . A coordinate vertex on $V(F)$ has stabilizer equal to the corresponding weight. On the open coordinate line joining variables of weights A_i, A_j , the stabilizer is $D = \gcd(A_i, A_j)$. If the restriction of F to that line is a binomial and its exponent vectors differ by $s(A_j/D, -A_i/D)$, then the line contributes $|s|$ points of order D . Orders one are omitted.

If $\mathcal{C}$ has exactly three nontrivial stacky points of orders r_1, r_2, r_3 , and $\delta = h - A - B - C$, weighted adjunction gives

$$(4.4) \quad 2g(\mathcal{C}) = \frac{\delta h}{ABC} - 1 + \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}.$$

Proof. The stabilizer assertions follow from the quotient presentation. After dividing the two weights by D , the effective action on the open line has coprime weights. Its invariant torus coordinate identifies the binomial with a nonzero scalar times $u^{|s|} - \lambda$. Quasismoothness makes its roots distinct, so there are $|s|$ orbits, each with residual stabilizer μ_D .

Also $\deg \mathcal{O}_{\mathcal{C}}(1) = h/(ABC)$ and $\omega_{\mathcal{C}} = \mathcal{O}_{\mathcal{C}}(\delta)$. Hence

$$\frac{\delta h}{ABC} = 2g(\mathcal{C}) - 2 + \sum_{i=1}^3 \left(1 - \frac{1}{r_i}\right),$$

which is equivalent to (4.4). $\square$

Lemma 4.6 (The seven ternary graph choices). *Let $F(x, y, z)$ be weighted homogeneous with an isolated critical point and no mixed quadratic summand. Up to a permutation, every choice in the ternary graph classification has one of the seven skeletons in Table 1. Rows 1, 2, 4, 5, 7 contain an atomic invertible subpolynomial of type FFF, C2F, L2F, C3, L3. In rows 3 and 6, either there is another unbranched atomic choice, or the same weights and degree admit a nondegenerate four-term polynomial obtained by adding a link $x^m z^n$ with $m, n > 0$.*

Proof. The list is classical [1, Section 13.2]; we use the convenient formulation in [19, Theorem 10], whose support criterion [19, Theorem 8] may be compared with [12]. Rows 1, 2, 4, 5, 7 visibly give the five disjoint unions of Fermat, chain, and loop atoms. We spell out the support check for the remaining rows.

row	graph skeleton	outcome
1	$x^a + y^b + z^c$	FFF
2	$x^a z + y^b + z^c$	C2F
3	$x^a y + z^c y + y^b$	branched f_3
4	$y^b + x^a z + z^c x$	L2F
5	$x^a z + z^c y + y^b$	C3
6	$x^a y + y^b z + z^c y$	branched f_6
7	$x^a y + y^b z + z^c x$	L3

TABLE 1. The seven graph choices in three variables. Nonzero coefficients are suppressed; atomic labels are understood up to permutation.

For row 3, take

$$\mathcal{R}_3 = \{x^a y, z^c y, y^b, x^m z^n\}, \quad m, n > 0.$$

The derivatives with respect to y of $x^a y, z^c y$ handle the singleton sets $\{x\}, \{z\}$, while y^b handles $\{y\}$. For $J = \{x, y\}$ and $J = \{y, z\}$, the term y^b verifies (C1), and the corresponding two derivatives verify (C2). For $J = \{x, z\}$, the link lies in $\mathbb{C}[x, z]$, and its two derivatives verify (C2). Hence $\mathcal{R}_3$ satisfies (C1)–(C2)'.

For row 6, put

$$\mathcal{R}_6 = \{x^a y, y^b z, z^c y, x^m z^n\}.$$

The singleton conditions follow from $\partial_y(x^a y) = x^a$, $\partial_z(y^b z) = y^b$, and $\partial_y(z^c y) = z^c$. For $J = \{x, y\}$ use $x^a y$ for (C1) and the derivatives with respect to x, z of $x^a y, y^b z$ for (C2); for $J = \{y, z\}$ use $y^b z$ and the derivatives with respect to y, z of $z^c y, y^b z$. The link handles $J = \{x, z\}$. Thus $\mathcal{R}_6$ also satisfies (C1)–(C2)', and [19, Theorem 8] makes a generic member of either four-term support nondegenerate.

For an arbitrary polynomial with row 3 or 6, condition (C1) for $J = \{x, z\}$ cannot use two indices outside J , since the complement has only one element. It therefore supplies a monomial in $\mathbb{C}[x, z]$. If one exponent is zero, a pure power produces an unbranched atomic choice; otherwise it is the positive link above. $\square$

Proposition 4.7 (The two branched ternary types are not triangles). *No quasismooth member of either noninvertible branched weight system in the ternary graph classification can have coarse genus zero and exactly three nontrivial stacky points.*

Proof. By Lemma 4.6, a genuinely branched weight system has, after permuting variables, a nondegenerate four-term member

$$(4.5) \quad f_3 = x^a y + z^c y + y^b + \epsilon x^m z^n,$$

$$(4.6) \quad f_6 = x^a y + y^b z + z^c y + \epsilon x^m z^n,$$

where $a, b, c \geq 2$, $m, n > 0$, and $\epsilon \neq 0$. For a weighted-homogeneous polynomial, nondegeneracy is equivalent to smoothness of the punctured cone, hence to quasismoothness of the stacky Proj. If a quasismooth polynomial of the same weights and degree were a polarized triangle, Lemma 4.4 would make this four-term member the same polarized triangle. It is therefore enough to compute the stacky points and coarse genus of the displayed members.

For (4.5), put $e = b - 1$ and

$$L = \text{lcm}(a, e, c), \quad (A, B, C) = \left(\frac{L}{a}, \frac{L}{e}, \frac{L}{c}\right), \quad h = L + B,$$

and set

$$r = \gcd(a, e), \quad s = \gcd(c, e),$$

$$U = \gcd(A, B) = \frac{L}{\text{lcm}(a, e)}, \quad V = \gcd(B, C) = \frac{L}{\text{lcm}(c, e)}.$$

The two coordinate vertices on the curve have orders A, C . The lines $z = 0$ and $x = 0$ contribute respectively r points of order U and s points of order V . Thus the number of nontrivial stacky points is

$$(4.7) \quad [A > 1] + [C > 1] + r[U > 1] + s[V > 1],$$

where brackets denote indicator functions.

The two numbers U and V cannot both exceed one, because then the count is at least four. If neither exceeds one, the count is at most two. Thus, up to $x \leftrightarrow z$, one has $U > 1$ and $V = 1$. Since $U > 1$ implies $A > 1$, equality with three leaves exactly

$$(I) \quad A > 1, \quad C = 1, \quad U > 1, \quad V = 1, \quad r = 2,$$

$$(II) \quad A > 1, \quad C > 1, \quad U > 1, \quad V = 1, \quad r = 1.$$

In case (II), the coprimality $r = 1$ gives $A = eU$ and $B = aU$. Since $V = \gcd(B, C) = 1$ and $cC = eB$, one has $C \mid e$, and hence $C \mid A$. On the other hand

$$h = (e + 1)B \not\equiv 0 \pmod{C},$$

because $C \mid e$ and $\gcd(B, C) = 1$. This contradicts the homogeneity equation $mA + nC = h$ for the link monomial.

In case (I), write $a = 2a_0$ and $e = 2e_0$, with $\gcd(a_0, e_0) = 1$. Since $C = 1$,

$$L = c = 2a_0e_0U, \quad (A, B, C) = (e_0U, a_0U, 1), \quad h = a_0U(2e_0 + 1).$$

The three orders are (e_0U, U, U) . Substitution in (4.4) gives

$$(4.8) \quad 2g = 2(2a_0e_0 + a_0 - e_0 - 1) = 2((a_0 - 1)(2e_0 + 1) + e_0) > 0.$$

This excludes f_3 .

For (4.6), put $e = b - 1$, $t = c - 1$, and

$$(4.9) \quad G = \gcd(e(t + 1), at, ae), \quad (A, B, C; h) = \frac{1}{G}(e(t + 1), at, ae; a(et + e + t)).$$

Let $d = \gcd(e, t)$ and $W = ad/G$. This is an integer because $G \mid \gcd(at, ae) = ad$. All three coordinate vertices lie on the curve, and the open line $x = 0$ contributes d points of order $\gcd(B, C) = W$. Hence the point count is

$$(4.10) \quad [A > 1] + [B > 1] + [C > 1] + d[W > 1].$$

If $W = 1$, equality with three forces $A, B, C > 1$. Write $e = de_0$, $t = dt_0$, where $\gcd(e_0, t_0) = 1$. Since $G = ad$,

$$(A, B, C; h) = \left(\frac{e_0(dt_0 + 1)}{a}, t_0, e_0; de_0t_0 + e_0 + t_0 \right).$$

Using the three vertex orders in (4.4) gives

$$(4.11) \quad 2g = d(a - 1) + \frac{a}{e_0} - 1 > 0.$$

Indeed $d(a - 1) \geq 1$ and $a/e_0 > 0$.

If $W > 1$, then $B, C > 1$, so (4.10) equals three only if $d = 1$ and $A = 1$. Formula (4.9) then gives $G = e(t + 1)$ and $a = We(t + 1)$, whence

$$(A, B, C; h) = (1, Wt, We; W(et + e + t)).$$

The orders are (Wt, We, W) , and adjunction gives

$$(4.12) \quad 2g = et + e + t - 1 > 0.$$

This excludes f_6 . □

Lemma 4.8 (Coordinate-stratum audit for atomic curves). *Let g be one of the five atomic polynomials with exponents $a, b, c \geq 2$, let $(A_0, B_0, C_0; h_0)$ be its adjugate weight data, and put $q = \gcd(A_0, B_0, C_0)$. Give g the primitive weights $(A, B, C; h) = (A_0, B_0, C_0; h_0)/q$. Table 2 lists every nontrivial stacky point of $[(V(g) \setminus \{0\})/\mathbb{G}_m]$. Here $V_x(D)$ denotes the x -vertex of order D , and $L_{ij}(d, D)$ denotes d points of order D on the open ij -line; entries of order one are omitted. If $q = 1$, the curve has exactly three stacky points, with orders given in (4.13).*

type	complete list of coordinate strata
FFF	$L_{xy}(d_{ab}, cd_{ab}/q), L_{xz}(d_{ac}, bd_{ac}/q), L_{yz}(d_{bc}, ad_{bc}/q), \quad d_{ab} = \gcd(a, b), \text{ cyclically}$
C2F	$V_x(c(b-1)/q), L_{xy}(r, cr/q), L_{yz}(s, as/q), \quad r = \gcd(a, b-1), \quad s = \gcd(b, c)$
L2F	$V_x(c(b-1)/q), V_y(c(a-1)/q), L_{xy}(d, cd/q), \quad d = \gcd(a-1, b-1)$
C3	$V_x((c(b-1)+1)/q), V_y(a(c-1)/q), L_{yz}(d, ad/q), \quad d = \gcd(b, c-1)$
L3	$V_x((c(b-1)+1)/q), V_y((a(c-1)+1)/q), V_z((b(a-1)+1)/q)$

TABLE 2. All stacky coordinate strata of the five atomic curves.

For $q = 1$, the resulting signatures are

$$(4.13) \quad \begin{array}{l|l} \text{FFF} & (a, b, c) \\ \text{C2F} & (a, c(b-1), c) \\ \text{L2F} & (c(a-1), c(b-1), c) \\ \text{C3} & (a, c(b-1)+1, a(c-1)) \\ \text{L3} & (b(a-1)+1, c(b-1)+1, a(c-1)+1). \end{array}$$

Proof. Restrict the atomic polynomial to each coordinate vertex and open coordinate line and apply Lemma 4.5. For example, the C2F restriction to the open xy -line is $y(x^a + y^{b-1})$; it has $\gcd(a, b-1)$ orbits, each of order $c \gcd(a, b-1)/q$. Its open yz -line gives $\gcd(b, c)$ orbits of order $a \gcd(b, c)/q$, and its x -vertex has order $c(b-1)/q$. The same binomial computation gives every entry in the table. On each unlisted open line the restriction is a single nonzero monomial, so no point of the curve lies there; the vertex list follows by checking for a pure power of the corresponding variable. Thus the table is exhaustive. On the dense torus the stabilizer is $\mu_{\gcd(A, B, C)} = \{1\}$, so there is no additional generic stratum.

When $q = 1$, primitivity in FFF is equivalent to pairwise coprimality of a, b, c . In C2F, both $\gcd(a, b-1)$ and $\gcd(b, c)$ divide all three raw weights; in L2F, $\gcd(a-1, b-1)$ does so; and in C3, $\gcd(b, c-1)$ does so. They are therefore one. Type L3 already has exactly its three vertices. Substitution gives (4.13), whose displayed orders are all at least two. □

Lemma 4.9 (Primitive atomic mutation). *Let g be a three-variable atomic invertible polynomial with primitive weights $(A, B, C; h)$ and $h - A - B - C > 0$. Suppose its stacky Proj has coarse genus zero and exactly three nontrivial stacky points. Then g is graded-equivalent to an atomic polynomial g' whose exponent determinant satisfies $|\det E_{g'}| = h$.*

Proof. The exponent matrix of an atomic polynomial is nonsingular. Hence the diagonal change-of-variables map $(\mathbb{C}^*)^3 \rightarrow (\mathbb{C}^*)^3$ on its three monomial coefficients is an isogeny, and we may normalize all three coefficients to one before beginning the case analysis.

Up to permutation, the five atomic forms and their unnormalized weight data are as follows:

type	polynomial	$(A_0, B_0, C_0; h_0)$
FFF	$x^a + y^b + z^c$	$(bc, ac, ab; abc)$
C2F	$x^a y + y^b + z^c$	$(c(b-1), ac, ab; abc)$
L2F	$x^a y + x y^b + z^c$	$(c(b-1), c(a-1), ab-1; c(ab-1))$
C3	$x^a y + y^b z + z^c$	$(c(b-1)+1, a(c-1), ab; abc)$
L3	$x^a y + y^b z + z^c x$	$(c(b-1)+1, a(c-1)+1, b(a-1)+1; abc+1)$

The complete stacky-point list used in the case analysis is Table 2. It includes all coordinate vertices and all zeroes on open coordinate lines; every unlisted restriction is a nonzero monomial. Thus the following point counts are exhaustive. Put $q = \gcd(A_0, B_0, C_0)$. Then $(A, B, C; h) = (A_0, B_0, C_0; h_0)/q$, and

$$(4.14) \quad |\det E_g| = h_0 = qh.$$

We prove that either $q = 1$ or g belongs to one of four explicitly mutable families.

For FFF, put $d_{ab} = \gcd(a, b)$ and $D_{ab} = cd_{ab}/q$, and define d_{ac}, D_{ac} and d_{bc}, D_{bc} cyclically. The line $z = 0$ contributes d_{ab} points precisely when $D_{ab} > 1$. Thus

$$\sum_{D_{ij} > 1} d_{ij} = 3,$$

and adjunction simplifies to

$$(4.15) \quad 2g = q \left(1 - \frac{1}{a} - \frac{1}{b} - \frac{1}{c} + \sum_{D_{ab} > 1} \frac{1}{c} \right) - 1,$$

where the last sum is cyclic over active pairs. If all three pairs are active, (4.15) is $q - 1$, so genus zero gives $q = 1$. If two pairs are active, their contributions are $2 + 1$. Relabel so that $d_{ab} = 2$, $d_{ac} = 1$, and $D_{bc} = 1$. Then $0 = 2g = q(1 - 1/a) - 1$. For the integer $q > 1$, this forces $a = q = 2$. It follows that $b = 2k$ and $1 = D_{bc} = \gcd(b, c)$; hence

$$(4.16) \quad g = x^2 + y^{2k} + z^\ell, \quad \gcd(2k, \ell) = 1, \quad q = 2.$$

If only one pair is active, its contribution is three. Say $d_{ab} = 3$. Now $0 = 2g = q(1 - 1/a - 1/b) - 1$. Since q is divisible by $\gcd(a, b) = 3$ and $a, b \geq 3$, the left factor is at least three, whereas $(1 - 1/a - 1/b)^{-1} \leq 3$. Equality forces

$$(4.17) \quad g = x^3 + y^3 + z^\ell, \quad \gcd(3, \ell) = 1, \quad q = 3.$$

For C2F, set

$$e = b - 1, \quad r = \gcd(a, e), \quad s = \gcd(b, c), \quad U = \frac{cr}{q}, \quad V = \frac{as}{q},$$

and let

$$X = [ce/q > 1], \quad Y = [U > 1], \quad Z = [V > 1].$$

The vertex x , the line $z = 0$, and the line $x = 0$ give respectively X, rY, sZ stacky points. In particular $Y = 1$ implies $X = 1$. Direct substitution of the raw weights in adjunction gives

$$(4.18) \quad 2g = q \left(1 - \frac{1-Z}{a} - \frac{1-Y}{c} - \frac{1-X}{ce} \right) - 1, \quad X + rY + sZ = 3.$$

If $Y = Z = 1$, then $X = r = s = 1$ and $2g = q - 1$, so $q = 1$. If $Y = 1$ and $Z = 0$, then $X = 1$, $r = 2$, and $0 = 2g = q(1 - 1/a) - 1$. Thus $a = q = 2$; moreover $e = 2k$ and $1 = V = s$. This is

$$(4.19) \quad g = x^2 y + y^{2k+1} + z^\ell, \quad \gcd(2k+1, \ell) = 1, \quad q = 2.$$

If $Y = 0$, $Z = 1$, and $X = 1$, then $s = 2$ and $0 = 2g = q(1 - 1/c) - 1$. Thus $c = q = 2$, while $1 = U = r$, giving

$$(4.20) \quad g = x^\alpha y + y^{2k} + z^2, \quad \gcd(\alpha, 2k - 1) = 1, \quad q = 2.$$

The only remaining count is $X = Y = 0$, $Z = 1$, and $s = 3$. Here $q = ce$, and (4.18) gives $2g = e(c - 1) - 2$. Genus zero would give $e(c - 1) = 2$; its two solutions $(e, c) = (1, 3), (2, 2)$ both have $\gcd(e + 1, c) = 1$, contrary to $s = 3$. This completes C2F.

For L2F, set $d = \gcd(a - 1, b - 1)$ and $D = cd/q$. Its two coordinate vertices and its open xy -line give

$$[c(b - 1)/q > 1] + [c(a - 1)/q > 1] + d[D > 1]$$

points. If $D = 1$ the count is at most two; if $D > 1$, both vertices occur, so a count of three forces $d = 1$. The three orders are then $c(b - 1)/q, c(a - 1)/q, c/q$, and substitution gives

$$(4.21) \quad 2g = q - 1.$$

Thus $q = 1$.

For C3, put $p = c(b - 1) + 1$, $d = \gcd(b, c - 1)$, and $D = ad/q$. The two possible vertices and the open yz -line give

$$[p/q > 1] + [a(c - 1)/q > 1] + d[D > 1]$$

points. The line must be active, and then the second vertex is present. If the first vertex is present, the count forces $d = 1$; the three orders are $p/q, a(c - 1)/q, a/q$, and adjunction again gives $2g = q - 1$. If the first vertex is absent, then $p/q = 1$ and the count forces $d = 2$. In that case substitution gives $2g = q - 2$. But $q = p = c(b - 1) + 1 \geq 4$ when $d = 2$, so this is positive. Hence a genus-zero triangle has $q = 1$.

Finally, L3 has only its three coordinate vertices. A three-point fiber therefore has orders $A_0/q, B_0/q, C_0/q$, and direct substitution gives

$$(4.22) \quad 2g = q - 1.$$

Thus $q = 1$ here as well. This proves that all cases with $q > 1$ are exactly (4.16), (4.17), (4.19), and (4.20), up to permutation.

It remains to remove their common factor. Choose $\lambda^2 = -1$. Completing a square, followed by a nonzero rescaling, gives graded changes of variables

$$\begin{aligned} x^2 + y^{2k} &\sim X^2 + Xy^k, \\ x^2y + y^{2k+1} &\sim X^2y + Xy^{k+1}, \\ y^{2k} + z^2 &\sim y^kZ + Z^2. \end{aligned}$$

Likewise a linear change of the equal-weight variables sends the binary cubic $x^3 + y^3$, whose three roots on $\mathbb{P}^1$ are distinct, to a nonzero multiple of $XY(X + Y)$. Thus the four replacements are

source	primitive atomic replacement
$x^2 + y^{2k} + z^\ell$	$X^2 + Xy^k + z^\ell$
$x^3 + y^3 + z^\ell$	$X^2Y + XY^2 + z^\ell$
$x^2y + y^{2k+1} + z^\ell$	$X^2y + Xy^{k+1} + z^\ell$
$x^\alpha y + y^{2k} + z^2$	$x^\alpha y + y^kZ + Z^2$

The gcds of the replacement adjugate weights are, in order,

$$\gcd(\ell, 2k), \quad \gcd(\ell, 3), \quad \gcd(\ell, 2k + 1), \quad \gcd(\alpha, 2k - 1),$$

so all four are one by the conditions above. Their exponent determinants are therefore their primitive degree h , by (4.14). In the first and fourth rows, $k = 1$ would give $h - A - B - C < 0$; hence the positivity hypothesis ensures that every displayed replacement has atomic exponents at least two. $\square$

Theorem 4.10 (Automatic Cox-invertibility in three variables). *Every three-variable fractional triangle hypersurface is Cox-invertible in the sense of Definition 4.1.*

Proof. The polarization has positive orbifold degree. Orbifold Riemann–Roch therefore gives $R_m \neq 0$ for every sufficiently large m . Thus the grading is effective: if the degrees of a homogeneous generating set had a common divisor greater than one, all nonzero graded pieces would have degree divisible by it. We may consequently take a primitive minimal presentation

$$R \cong \mathbb{C}[x, y, z]/(F), \quad \deg(x, y, z; F) = (A, B, C; h).$$

Since R is normal of dimension two, the singular locus of its affine cone has codimension at least two and is therefore zero-dimensional. The weighted Euler identity $hF = Ax F_x + By F_y + Cz F_z$ shows that the critical locus is precisely the cone singular locus. It is $\mathbb{G}_m$ -invariant, while any nonzero critical point has a one-dimensional orbit because all weights are positive and its stabilizer is finite. This contradicts zero-dimensionality. Thus the vertex is the only critical point and F is nondegenerate.

We first dispose of the quadratic case excluded from the graph convention in [19]. If a mixed quadratic term xy occurred, then $A + B = h$, so the positive a -invariant identity would give $h - A - B - C = -C < 0$, a contradiction. Pure squares are Fermat root monomials and are already among the graph choices. Thus the ternary graph classification applies without invoking a separate splitting lemma.

Lemma 4.6 leaves five unbranched atomic types and the two branched types f_3, f_6 . Proposition 4.7, together with deformation rigidity, excludes the latter two. Hence F has a nondegenerate atomic invertible skeleton g of the same weights and degree.

Lemma 4.2 identifies the polarized stacky Proj of F with the original three-point Geigle–Lenzing curve. Both F and g lie in the quasismooth locus for this fixed weight system. Lemma 4.4 identifies $(\text{Proj}_{\text{st}} R, \mathcal{O}(1))$ with the polarized stacky Proj of g . Lemma 4.9 replaces g by a graded-equivalent primitive atomic polynomial g' with $|\det E_{g'}| = h$.

It remains to recover the affine cone. For every $m \in \mathbb{Z}$, the degree- m functions on the punctured cone are $H^0(\text{Proj}_{\text{st}} R, \mathcal{O}(m))$. Normality and dimension two give Hartogs extension across the vertex, equivalently $H_{R_+}^0(R) = H_{R_+}^1(R) = 0$. Hence

$$R_m = H^0(\text{Proj}_{\text{st}} R, \mathcal{O}(m)) \quad (m \geq 0),$$

and therefore

$$R \cong \bigoplus_{m \geq 0} H^0(\text{Proj}_{\text{st}} R, \mathcal{O}(m)) \cong \mathbb{C}[x, y, z]/(g')$$

as graded rings. Proposition 5.5 now supplies the Cox monomial lift. To make the passage back to the original Cox ring explicit, the polarized stack isomorphism identifies the three stabilizer orders of g' with the original signature and the stabilizer characters of $\mathcal{O}(1)$ with those of $\mathcal{O}(\vec{\tau})$, up to the same permutation. Theorem 1.1 then identifies the root class itself, since the δ -th root is unique. Because the exponent matrix of g' is nonsingular, a diagonal change of its three generators normalizes its three nonzero coefficients; hence we may take g' in the literal atomic normal form displayed in Table 3. Apply the appropriate row of Table 4 directly to the original variables X, Y, Z , ordered by these three stacky points. The displayed $L(p, q, r)$ degree identities in Proposition 5.5 give $\deg M_i = a_i \vec{\tau}$ and $\delta \vec{\tau} = \vec{\omega}$, while its Cox identity pulls the atomic relation back to a common monomial times $X^p + Y^q + Z^r$. This gives a specific graded homomorphism

$$\Psi: \mathbb{C}[x, y, z]/(g') \longrightarrow R(p, q, r; \vec{\tau})$$

in the original Cox coordinates. The injectivity and equal-Hilbert-function argument of Proposition 5.5 applies verbatim to Ψ : the monomial torus map is finite with dense image on the hypersurface, and the polarized stack identification gives equality of Hilbert functions. Hence Ψ is an isomorphism. Thus no unproved lifting of an abstract stack automorphism or rescaling of the fixed Brieskorn–Pham Cox coordinates is being used, and R is Cox-invertible. $\square$

Corollary 4.11 (Unmarked classification by primitive weights). *Let two fractional triangle hypersurfaces have the same sorted primitive weight system $(a, b, c; h)$. Then they are isomorphic as graded rings. Consequently the census counts unmarked graded-isomorphism classes, while distinct atomic parameter triples with the same weight key are merely distinct marked witnesses for one class.*

Proof. Both equations lie in the irreducible quasismooth locus of degree- h forms of weights (a, b, c) . Lemma 4.4 identifies their polarized stacky Proj. Normality, dimension two, and the depth-two local-cohomology sequence used above reconstruct each graded ring as

$$\bigoplus_{m \geq 0} H^0(\text{Proj}_{\text{st}} R, \mathcal{O}(m)).$$

Thus the polarized-stack isomorphism induces a graded-ring isomorphism. $\square$

5. ATOMIC MODELS AND DIVISOR CLASSIFICATION

Proposition 5.1 (Determinant and diagonal symmetry). *Let $f = \sum_{i=1}^3 \prod_{j=1}^3 z_j^{A_{ij}}$ be an invertible polynomial with primitive weights*

$$A(a, b, c)^T = h(1, 1, 1)^T, \quad \gcd(a, b, c) = 1.$$

Let

$$G_f = \left\{ (\lambda_1, \lambda_2, \lambda_3) \in (\mathbb{C}^*)^3 : \prod_j \lambda_j^{A_{ij}} = 1 \text{ for all } i \right\}$$

and let $J_f = (e^{2\pi i a/h}, e^{2\pi i b/h}, e^{2\pi i c/h})$ be the exponential grading element. Then

$$|G_f| = |\det A|, \quad \text{ord}(J_f) = h,$$

and hence

$$(5.1) \quad |\det A| = h \iff G_f = \langle J_f \rangle.$$

Thus the determinant condition says exactly that the residual diagonal symmetry group $G_f / \langle J_f \rangle$ is trivial.

Proof. On character lattices, the homomorphism of tori whose kernel is G_f is represented by A^T , so $|G_f| = |\det A|$. Weighted homogeneity shows that $J_f \in G_f$. Moreover,

$$\text{ord}(J_f) = \text{lcm}_j \frac{h}{\gcd(h, a_j)} = \frac{h}{\gcd(h, a, b, c)} = h,$$

where $\gcd(h, a, b, c) = 1$ follows from the primitive weights. The subgroup $\langle J_f \rangle$ therefore has order h , proving (5.1). $\square$

Remark 5.2. In the canonical-weight notation of Ebeling–Takahashi [9, Sections 1–2, especially Propositions 3 and 5], our G_f is their G_f^{fin} and $\langle J_f \rangle$ is their $G_0 = \langle g_0 \rangle$. They identify $G_0^T \simeq \text{Hom}(G_f^{\text{fin}}/G_0, \mathbb{C}^*)$ and prove $|G_0^T| = |G_f^{\text{fin}}/G_0| = |G_f/\langle J_f \rangle| = c_f$. Thus (5.1) is the familiar condition $c_f = 1$; the proposition is included only to translate that convention into the primitive grading used here.

The determinant condition concerns the existence of a suitable minimal presentation, not an arbitrary equation of the same graded ring.

Example 5.3. The primitive weight system $(12, 8, 3; 24)$ admits the Fermat equation $x^2 + y^3 + z^8$, whose exponent determinant is 48. Completing the square gives a graded-equivalent equation of the form $x^2 + xz^4 + y^3$, whose exponent determinant is 24. Thus determinant equals degree is an existence property of a minimal presentation and must not be asserted for every presentation.

The ring R is a normal positively graded surface with $R_0 = \mathbb{C}$. Its singular locus is $\mathbb{G}_m$ -stable; normality excludes a one-dimensional singular locus, so a hypersurface presentation has an isolated critical point at its vertex. Minimality excludes a linear monomial. A mixed quadratic is impossible by the positive a -invariant identity used in the proof of Theorem 4.10, while pure squares are Fermat atoms. Consequently the atomic classification applies: an isolated three-monomial quasihomogeneous polynomial is, up to permutations, a disjoint union of Fermat, chain, and loop atoms [13]. In three variables there are five possibilities. The next table records the integral adjugate weights before common-factor normalization. The minimal determinant condition retains a row only when these data are primitive. We use $\alpha, \beta, \gamma \geq 2$ for exponents.

Type	Polynomial	Weight data and equation for δ
FFF	$x^\alpha + y^\beta + z^\gamma$	$(a, b, c; h) = (\beta\gamma, \alpha\gamma, \alpha\beta; \alpha\beta\gamma),$ $\delta = \alpha\beta\gamma - \alpha\beta - \alpha\gamma - \beta\gamma$
C2F	$x^\alpha y + y^\beta + z^\gamma$	$(a, b, c; h) = (\gamma(\beta - 1), \alpha\gamma, \alpha\beta; \alpha\beta\gamma),$ $\gamma(\alpha - 1)(\beta - 1) = \alpha\beta + \delta$ $(a, b, c; h) = (\gamma(\beta - 1), \gamma(\alpha - 1), \alpha\beta - 1;$ $\gamma(\alpha\beta - 1)),$
L2F	$x^\alpha y + xy^\beta + z^\gamma$	$\gamma(\alpha - 1)(\beta - 1) = \alpha\beta - 1 + \delta$ $(a, b, c; h) = (\beta\gamma - \gamma + 1, \alpha(\gamma - 1), \alpha\beta;$ $\alpha\beta\gamma),$
C3	$x^\alpha y + y^\beta z + z^\gamma$	$\gamma(\alpha - 1)(\beta - 1) = \alpha\beta - \alpha + 1 + \delta$ $(a, b, c; h) = (\beta\gamma - \gamma + 1, \alpha\gamma - \alpha + 1,$ $\alpha\beta - \beta + 1; \alpha\beta\gamma + 1),$
L3	$x^\alpha y + y^\beta z + z^\gamma x$	$\gamma(\alpha - 1)(\beta - 1) = \alpha\beta - \alpha - \beta + 2 + \delta$

TABLE 3. Atomic equations for a minimal invertible presentation.

The equations in Table 3 are obtained simply by imposing $h - a - b - c = \delta$. The following elementary bound turns them into a provably finite enumeration.

Lemma 5.4. *For a primitive solution in Table 3, all three exponents are at most $\delta + 6$, after permuting the exponents in type FFF.*

Proof. Put $A = \alpha - 1$ and $B = \beta - 1$. For the last four types, the displayed equations become, respectively,

$$\begin{aligned} (\gamma - 1)AB &= A + B + \delta + 1, & (\gamma - 1)AB &= A + B + \delta, \\ (\gamma - 1)AB &= B + \delta + 1, & (\gamma - 1)AB &= \delta + 1. \end{aligned}$$

In the first two equations one obtains $(A - 1)(B - 1) \leq \delta + 2$ and $(A - 1)(B - 1) \leq \delta + 1$, respectively. If $A = 1$ or $B = 1$, the remaining factor divides the corresponding right-hand constant. In the third equation $(A - 1)B \leq \delta + 1$, with the same divisibility conclusion when $A = 1$; the fourth says directly that $AB \mid \delta + 1$. These inequalities bound α and β by $\delta + 4$, while the four displayed formulae bound γ by $\delta + 4$.

For FFF order $\alpha \leq \beta \leq \gamma$. If $\alpha = 2$, its equation is

$$(\beta - 2)(\gamma - 2) = \delta + 4,$$

so the assertion follows by divisibility. If $\alpha, \beta \geq 3$, then

$$\gamma = \frac{\alpha\beta + \delta}{\alpha\beta - \alpha - \beta} \leq 3 + \frac{\delta}{3} \leq \delta + 3;$$

here $\alpha\beta/(\alpha\beta - \alpha - \beta) \leq 3$. This also bounds α and β because of the chosen order. $\square$

Proposition 5.5 (Primitive atomic data are Cox data). *Let one of the five atomic polynomials in Table 3 have primitive adjugate weights and $h - a - b - c = \delta > 0$. Then its graded hypersurface is a Cox-invertible fractional triangle ring of index δ . Putting $A = \alpha - 1$, $B = \beta - 1$, and $t = \gamma - 1$, its signature and Cox lift are given by Table 4. Conversely, every Cox-invertible fractional triangle hypersurface arises from primitive atomic data in this way.*

Type	(p, q, r)	(M_1, M_2, M_3)	M_0
FFF	(α, β, γ)	(X, Y, Z)	1
C2F	$(\alpha, \gamma B, \gamma)$	(X, Y^γ, YZ)	Y^γ
L2F	$(\gamma A, \gamma B, \gamma)$	$(X^\gamma, Y^\gamma, XYZ)$	$X^\gamma Y^\gamma$
C3	$(\alpha, \gamma B + 1, \alpha t)$	$(XZ, Y^\gamma, YZ^\alpha)$	$Y^\gamma Z^\alpha$
L3	$(\beta A + 1, \gamma B + 1, \alpha t + 1)$	$(X^\beta Z, XY^\gamma, YZ^\alpha)$	$X^\beta Y^\gamma Z^\alpha$

TABLE 4. Signatures and Cox lifts attached to primitive atomic data.

Proof. Let $S = \mathbb{C}[x, y, z]/(f)$ with its primitive adjugate grading and set

$$\mathcal{C} = [(\text{Spec } S \setminus \{0\})/\mathbb{G}_m].$$

The polynomial is invertible and nondegenerate, hence has an isolated critical point. Therefore S is a normal surface and $\mathcal{C}$ is a smooth stacky curve. Primitivity makes its generic stabilizer trivial. Lemma 4.8, with normalization factor equal to one, proves that $\mathcal{C}$ has exactly three stacky points and gives their orders as Table 4. At each such point the stabilizer is a subgroup of the grading $\mathbb{G}_m$, so its action on $\mathcal{O}_{\mathcal{C}}(1)$ is the tautological, hence faithful, character. The coordinate-stratum table also verifies that there are no hidden vertex or open-line orbits.

Substitution in each row gives

$$(5.2) \quad \frac{\delta h}{abc} = 1 - \frac{1}{p} - \frac{1}{q} - \frac{1}{r} > 0.$$

Here $\deg \mathcal{O}_{\mathcal{C}}(1) = h/(abc)$, by weighted intersection theory or the leading coefficient of the hypersurface Hilbert series. Weighted adjunction gives $\omega_{\mathcal{C}} \cong \mathcal{O}_{\mathcal{C}}(\delta)$. Comparing (5.2) with

$$\deg \omega_{\mathcal{C}} = 2g(\mathcal{C}) - 2 + \sum_{s \in \{p, q, r\}} \left(1 - \frac{1}{s}\right)$$

gives $g(\mathcal{C}) = 0$. Hence $\mathcal{C} \cong \mathbb{X}(p, q, r)$, and under this isomorphism $\mathcal{O}_{\mathcal{C}}(1)$ is a line bundle $\mathcal{O}_{\mathbb{X}}(\vec{\tau})$ satisfying $\delta \vec{\tau} = \vec{\omega}$.

The four non-Fermat Cox identities are

$$\begin{aligned} X^\alpha Y^\gamma + Y^{\gamma\beta} + Y^\gamma Z^\gamma &= Y^\gamma (X^\alpha + Y^{\gamma B} + Z^\gamma), \\ X^{\gamma\alpha} Y^\gamma + X^\gamma Y^{\gamma\beta} + X^\gamma Y^\gamma Z^\gamma &= X^\gamma Y^\gamma (X^{\gamma A} + Y^{\gamma B} + Z^\gamma), \\ X^\alpha Y^\gamma Z^\alpha + Y^{\gamma\beta+1} Z^\alpha + Y^\gamma Z^{\alpha\gamma} &= Y^\gamma Z^\alpha (X^\alpha + Y^{\gamma B+1} + Z^{\alpha t}), \\ X^{\alpha\beta+1} Y^\gamma Z^\alpha + X^\beta Y^{\beta\gamma+1} Z^\alpha + X^\beta Y^\gamma Z^{\alpha\gamma+1} &= X^\beta Y^\gamma Z^\alpha (X^{\beta A+1} + Y^{\gamma B+1} + Z^{\alpha t+1}). \end{aligned}$$

They give the last four rows of the table; the FFF identity is tautological. For each row, the relations $p\vec{x} = q\vec{y} = r\vec{z} = \vec{c}$ in $L(p, q, r)$ give directly, writing $(a_1, a_2, a_3) = (a, b, c)$,

$$a_j \deg_L M_i = a_i \deg_L M_j, \quad \delta \deg_L M_i = a_i \vec{\omega}.$$

Choose integers v_i with $\sum_i v_i a_i = 1$ and put $\vec{\tau} = \sum_i v_i \deg_L M_i$. Then

$$\deg_L M_i = a_i \vec{\tau}, \quad \delta \vec{\tau} = \vec{\omega}.$$

Thus the Cox substitution defines a graded homomorphism

$$\Phi: S \longrightarrow R(\mathbf{p}, \vec{\tau}).$$

It is injective. Indeed, the exponent matrix of (M_1, M_2, M_3) has nonzero determinant, so the corresponding map of three-dimensional tori is finite. Both Laurent hypersurfaces are irreducible, since their coordinate rings are localizations of the normal domains T and S . The restriction of the finite torus map to the Brieskorn–Pham Laurent hypersurface is finite, hence has closed irreducible two-dimensional image. The Cox identity places that image in the irreducible two-dimensional atomic Laurent hypersurface, so the image is the whole hypersurface. The localized homomorphism is therefore injective. Since S is a domain, localization at xyz is injective; hence Φ itself is injective.

It remains to see that no sections are missing. Since S is a normal Cohen–Macaulay surface, $\text{depth}_{S_+} S = 2$, and the standard local-cohomology sequence gives

$$S_m \xrightarrow{\sim} H^0(\mathcal{C}, \mathcal{O}_{\mathcal{C}}(m)) \quad (m \geq 0).$$

The polarized identification $(\mathcal{C}, \mathcal{O}(1)) \cong (\mathbb{X}(p, q, r), \mathcal{O}(\vec{\tau}))$ therefore shows that S and $R(\mathbf{p}, \vec{\tau})$ have the same Hilbert function. The injective graded map Φ is consequently an isomorphism. In particular, it sends the three minimal generators to the specific monomials in Table 4. Conversely, the relation in Definition 4.1 is an isolated invertible three-monomial polynomial. The atomic classification gives one of the five rows of Table 3; the determinant condition is precisely primitivity of the adjugate weights. $\square$

The small-index exponent box can be sharpened to a divisor parametrization valid for every root index.

Theorem 5.6 (Divisor parametrization). *Fix $\delta > 0$ and put $A = \alpha - 1$, $B = \beta - 1$, $t = \gamma - 1$. All Cox-invertible fractional triangle hypersurfaces of index δ are obtained from the following finite constructions, followed by the primitivity test $\gcd(a, b, c) = 1$ and identification of repeated graded presentations.*

(FFF) For $2 \leq \alpha \leq \delta + 6$, factor

$$UV = A\delta + (A + 1)^2, \quad U \leq V, \quad U \equiv -1 \pmod{A},$$

and set

$$\beta = 1 + \frac{U + 1}{A}, \quad \gamma = 1 + \frac{V + 1}{A},$$

retaining $\alpha \leq \beta \leq \gamma$.

(C2F) For $1 \leq t \leq \delta + 3$, take every ordered positive factorization

$$UV = t(\delta + 1) + 1, \quad U \equiv -1 \pmod{t},$$

and set $A = (U + 1)/t$, $B = (V + 1)/t$.

(L2F) For $1 \leq t \leq \delta + 2$, factor

$$UV = t\delta + 1, \quad U \leq V, \quad U \equiv -1 \pmod{t},$$

and again set $A = (U + 1)/t$, $B = (V + 1)/t$.

(C3) Choose an ordered positive factorization $\delta + 1 = BU$, choose a positive divisor t of $U + 1$, and set $A = (U + 1)/t$.

(L3) Choose an ordered factorization $\delta + 1 = tAB$.

In the last four cases recover $(\alpha, \beta, \gamma) = (A + 1, B + 1, t + 1)$. The construction is exact and has no heuristic cutoff.

Proof. The last four equations in Table 3 become

$$(5.3) \quad tAB = A + B + \delta + 1, \quad tAB = A + B + \delta,$$

$$(5.4) \quad B(tA - 1) = \delta + 1, \quad tAB = \delta + 1.$$

Completing products in the first two gives

$$(tA - 1)(tB - 1) = t(\delta + 1) + 1, \quad (tA - 1)(tB - 1) = t\delta + 1,$$

respectively. Thus $U = tA - 1$ and $V = tB - 1$ give (C2F) and (L2F); the unlisted congruence for V follows because each right side is congruent to 1 modulo t . Equations (5.3)–(5.4) give the stated bounds on t and the constructions (C3) and (L3).

For FFF, order the exponents and set

$$U = A(\beta - 1) - 1, \quad V = A(\gamma - 1) - 1.$$

Expanding its equation in Table 3 gives $UV = A\delta + (A + 1)^2$. Conversely, the congruence and recovery formulas reverse the calculation; the congruence for V is automatic. The bound on α is Lemma 5.4. Proposition 5.5 proves that these and only these primitive data give the asserted rings. $\square$

The final primitivity test can itself be written without computing the gcd of three adjugate weights.

Proposition 5.7 (Closed primitivity criteria). *Put $A = \alpha - 1$, $B = \beta - 1$, and $t = \gamma - 1$. A parameter triple in one of the five rows of Table 3 has primitive adjugate weights if and only if the corresponding conditions below hold:*

type	primitivity criterion
FFF	$\gcd(A + 1, B + 1) = \gcd(B + 1, t + 1) = \gcd(t + 1, A + 1) = 1,$
C2F	$\gcd(A + 1, B) = \gcd(A + 1, t + 1) = \gcd(t + 1, B + 1) = 1,$
L2F	$\gcd(A, B) = \gcd(t + 1, AB + A + B) = 1,$
C3	$\gcd(A + 1, B(t + 1) + 1) = \gcd(t, B + 1) = 1,$
L3	$\gcd(tAB - 1, B(t + 1) + 1) = 1.$

Consequently an index δ is a nonexistence index precisely when every candidate in the five finite divisor constructions of Theorem 5.6 violates the corresponding row.

Proof. For FFF, a prime divides all three weights $\beta\gamma, \alpha\gamma, \alpha\beta$ exactly when it divides at least two of α, β, γ . For C2F the weights are

$$(B(t + 1), (A + 1)(t + 1), (A + 1)(B + 1)).$$

A common prime either divides $t + 1$, in which case it must also divide $A + 1$ or $B + 1$, or it divides both B and $A + 1$. For L2F the weights are

$$(B(t + 1), A(t + 1), AB + A + B).$$

A common prime not dividing $t + 1$ divides both A and B ; a common prime dividing $t + 1$ must divide $AB + A + B$. For C3 the weights are

$$(B(t + 1) + 1, t(A + 1), (A + 1)(B + 1)).$$

A common prime either divides $A + 1$ and the first entry, or divides both t and $B + 1$. These observations give the first four rows.

For L3 put $m = tAB = \delta + 1$ and

$$w_1 = B(t + 1) + 1, \quad w_2 = t(A + 1) + 1, \quad w_3 = A(B + 1) + 1.$$

The identities

$$Aw_1 = m - 1 + w_3, \quad Bw_2 = m - 1 + w_1, \quad tw_3 = m - 1 + w_2$$

and $\gcd(ABt, m - 1) = 1$ imply

$$\gcd(w_1, w_2, w_3) = \gcd(m - 1, w_1).$$

This is the last row. $\square$

Corollary 5.8 (Unrestricted classification). *The divisor parametrization of Theorem 5.6 and the tests of Proposition 5.7 classify all three-variable fractional triangle hypersurfaces for every index $\delta > 0$.*

Proof. Apply Theorem 4.10 and Proposition 5.5. $\square$

6. CONSEQUENCES AND LOW-INDEX DATA

Corollary 6.1. *If $6 \nmid \delta$, there exists a Cox-invertible fractional triangle hypersurface of root index δ .*

Proof. If $3 \nmid \delta$, take the L3 polynomial

$$x^2y + y^2z + z^{\delta+2}x$$

with primitive weights $(\delta + 3, 2\delta + 3, 3; 4\delta + 9)$ and signature $(3, \delta + 3, 2\delta + 3)$; their greatest common divisor is $\gcd(\delta + 3, 3) = 1$. If $\delta \equiv 3 \pmod{6}$, take L3 exponents $(2, 3, (\delta + 3)/2)$; the primitive weights and signature are, respectively,

$$(\delta + 4, \delta + 2, 4; 3\delta + 10), \quad (4, \delta + 4, \delta + 2).$$

Here $\delta + 2$ and $\delta + 4$ are odd and differ by 2, hence are coprime; therefore $\gcd(\delta + 4, \delta + 2, 4) = 1$. These two cases exhaust $6 \nmid \delta$. $\square$

Theorem 6.2 (Census through index thirty). *For $2 \leq \delta \leq 30$, the numbers of fractional triangle hypersurfaces are*

δ	2	3	4	5	6	7	8	9	10	11
#	6	8	8	21	0	29	11	12	6	37
δ	12	13	14	15	16	17	18	19	20	21
#	0	36	10	9	11	40	2	46	11	12
δ	22	23	24	25	26	27	28	29	30	
#	9	55	3	28	16	18	10	52	0	

In total there are 506 graded isomorphism classes in this range.

Proof. For every δ , enumerate the finite factorizations of Theorem 5.6 and apply the closed primitivity conditions of Proposition 5.7. A canonical key is the sorted primitive weight system together with the relation degree, $(a, b, c; h)$ with $a \leq b \leq c$. Corollary 4.11 shows that equal keys represent the same graded-isomorphism class. Conversely, a graded isomorphism preserves the Hilbert series, while a minimal three-generator hypersurface presentation recovers the multiset of generator degrees and the relation degree; hence distinct keys are not graded-isomorphic. Completeness follows from Corollary 5.8.

The certificate `fractional_triangles_v04_classes.csv` contains one row for every canonical key, including a canonical atomic witness, signature, and Cox lift. The file `fractional_triangles_v04_witness_to_class.tsv` maps all 652 retained marked witnesses to their classes. More precisely, 433 classes have one retained witness, while 73 L3 classes have the three cyclic choices of starting vertex, giving $433 + 3 \cdot 73 = 652$. This is the certificate convention described in the Introduction, not an intrinsic class count. Exact enumeration gives the displayed per-index counts and the computed total 506. $\square$

Remark 6.3 (The six systems absent from the arXiv-v1 comparison). The six systems in Remark 1.5 have the following weighted-homogeneous equations, in the same order:

$$\begin{aligned} & xz^2 + y^3 + x^5y, & zx^3 + yz^2 + y^4, \\ & z^2 + xy^4 + x^5y, & z^2 + xy^4 + zx^4, & xz^2 + y^4 + x^4z, & yz^2 + x^4z + xy^3. \end{aligned}$$

Their weights are respectively the six displayed systems, so weighted homogeneity is immediate. The first and sixth equations agree, up to the order of summands, with the models in [7, Table 3]; the second and third weight systems occur in [6, Table 2(a)]. The atomic Cox lifts in Table 4 identify their stacky Proj with the corresponding root triangle and prove normality.

Remark 6.4 (Automorphic interpretation). Uniformize $\mathbb{X}(p, q, r)$ by a cocompact triangle group. The root $\mathcal{O}(\bar{\tau})^\delta \simeq \omega_{\mathbb{X}}$ is equivalently a compatible lift to the appropriate finite central cover together with a multiplier system. Its degree- m sections are automorphic forms of fractional weight $2m/\delta$, and $R(p, q, r; \bar{\tau})$ is their graded ring. This is the three-point Cox realization of the higher-spin/Fuchsian-lift picture in [16]; spin and canonical rings of stacky curves are studied in [14, 23].

The classical three-generator triangle-group picture goes back at least to Milnor. As recalled in [23, Remark 10.2.4], Milnor [15, Section 6] constructs a fractional-weight automorphic-form ring generated by three forms satisfying a Fermat relation. Neumann [17] develops the related connection between Brieskorn complete intersections and automorphic forms. These results give the historical analytic source of the Fermat Cox relation; they do not provide the automatic converse, which starts from an arbitrary three-generator root hypersurface and forces a Cox-invertible normal form.

DATA AND CODE AVAILABILITY

The version-04 supplementary archive contains the unrestricted exact divisor enumerator, a complete 506-row canonical-class certificate, the map from all 652 atomic witness rows retained by the supplement’s explicit normal-form convention to those classes, the source-audited Watanabe comparison, and hardened regression tests for the automatic-Cox argument. As an independent census check, the verifier scans the five generic atomic exponent matrices through the proved bound $2 \leq \text{exponent} \leq 36$, computes adjugate weights by determinants and Cramer’s rule rather than the divisor formulas, and reproduces all 506 class keys and 652 retained witness rows for $2 \leq \delta \leq 30$. Every failed check raises an explicit exception and remains active under `python -0`. The README specifies the witness convention, software versions, reconstruction commands, source-PDF hash, and MIT license. The verifier records whether that hash was checked, the source was re-extracted, and tables were compared byte-for-byte; it also asserts that the source weights are sorted. The internal `SHA256SUMS` manifest names and authenticates exactly every other distributed top-level file. All calculations use deterministic integer arithmetic, with no random choices or numerical tolerances.

ACKNOWLEDGEMENTS

K. U. was partially supported by Grants-in-Aid for Scientific Research 15KT0105, 16K13743, and 16H03930.

REFERENCES

- [1] V. I. Arnol’d, S. M. Gusein-Zade, and A. N. Varchenko, *Singularities of Differentiable Maps, Vol. I*, Monographs in Mathematics 82, Birkhäuser, Boston, 1985.
- [2] D. Abramovich, M. Olsson, and A. Vistoli, *Tame stacks in positive characteristic*, Ann. Inst. Fourier (Grenoble) **58** (2008), 1057–1091.

- [3] C. Cadman, *Using stacks to impose tangency conditions on curves*, Amer. J. Math. **129** (2007), 405–427.
- [4] I. V. Dolgachev, *Automorphic forms and quasihomogeneous singularities*, Funct. Anal. Appl. **9** (1975), 149–151.
- [5] I. V. Dolgachev, *On the link space of a Gorenstein quasihomogeneous surface singularity*, Math. Ann. **265** (1983), 529–540.
- [6] W. Ebeling, *The Milnor lattices of the elliptic hypersurface singularities*, Proc. London Math. Soc. (3) **53** (1986), 85–111.
- [7] W. Ebeling, *Mirror symmetry, Kobayashi’s duality, and Saito’s duality*, Kodai Math. J. **29** (2006), 319–336.
- [8] W. Ebeling and A. Takahashi, *Strange duality of weighted homogeneous polynomials*, Compos. Math. **147** (2011), 1413–1433.
- [9] W. Ebeling and A. Takahashi, *Mirror symmetry between orbifold curves and cusp singularities with group action*, Int. Math. Res. Not. IMRN (2013), no. 10, 2240–2270.
- [10] W. Geigle and H. Lenzing, *A class of weighted projective curves arising in representation theory of finite-dimensional algebras*, in *Singularities, Representation of Algebras, and Vector Bundles*, Lecture Notes in Math. 1273, Springer, 1987, 265–297.
- [11] K. Hashimoto, H. Lee, and K. Ueda, *On a certain generalization of triangle singularities*, Manuscripta Math. **153** (2017), 35–51.
- [12] C. Hertling and R. Kurbel, *On the classification of quasihomogeneous singularities*, J. Singul. **4** (2012), 131–153.
- [13] M. Kreuzer and H. Skarke, *On the classification of quasihomogeneous functions*, Comm. Math. Phys. **150** (1992), 137–147.
- [14] A. Landesman, P. Ruhm, and R. Zhang, *Spin canonical rings of log stacky curves*, Ann. Inst. Fourier (Grenoble) **66** (2016), 2339–2383.
- [15] J. Milnor, *On the 3-dimensional Brieskorn manifolds $M(p, q, r)$* , in *Knots, Groups, and 3-Manifolds*, Ann. of Math. Stud. 84, Princeton Univ. Press, Princeton, NJ, 1975, 175–225.
- [16] S. Natanzon and A. Pratussevitch, *Topological invariants and moduli of Gorenstein singularities*, J. Singul. **7** (2013), 61–87.
- [17] W. D. Neumann, *Brieskorn complete intersections and automorphic forms*, Invent. Math. **42** (1977), 285–293.
- [18] E. M. O’Dorney, *Canonical rings of $\mathbb{Q}$ -divisors on $\mathbb{P}^1$* , Ann. Comb. **19** (2015), 765–784.
- [19] A. Rarovskii, *Non-invertible quasihomogeneous singularities and their Landau–Ginzburg orbifolds*, J. Singul. **28** (2025), 217–233.
- [20] K. Saito, *Regular systems of weights and their associated singularities*, in *Complex Analytic Singularities*, Adv. Stud. Pure Math. 8, North-Holland, Amsterdam, 1987, 479–526.
- [21] K. Saito, *Towards a categorical construction of Lie algebras*, in *Algebraic Geometry in East Asia—Hanoi 2005*, Adv. Stud. Pure Math. 50, Math. Soc. Japan, Tokyo, 2008, 101–175.
- [22] A. Takahashi, *Weighted projective lines associated to regular systems of weights of dual type*, in *New Developments in Algebraic Geometry, Integrable Systems and Mirror Symmetry*, Adv. Stud. Pure Math. 59, Math. Soc. Japan, Tokyo, 2010, 371–388.
- [23] J. Voight and D. Zureick-Brown, *The canonical ring of a stacky curve*, Mem. Amer. Math. Soc. **277** (2022), no. 1362.
- [24] P. Wagreich, *Algebras of automorphic forms with few generators*, Trans. Amer. Math. Soc. **262** (1980), 367–389.
- [25] K.-i. Watanabe, *Classification of 2-dimensional graded normal hypersurfaces with $a(R) \leq 6$* , Bull. Braz. Math. Soc. (N.S.) **45** (2014), 887–920.
- [26] K.-i. Watanabe, *Classification of 2-dimensional graded normal hypersurfaces with $a(R) \leq 6$* , arXiv:1401.0789v1 [math.AC], 4 January 2014, <https://arxiv.org/abs/1401.0789v1>.

KENJI HASHIMOTO, MAX PLANCK INSTITUTE FOR MATHEMATICS

Email address: hashimoto@mpim-bonn.mpg.de

HWAYOUNG LEE, DEPARTMENT OF MATHEMATICAL SCIENCES, SEOUL NATIONAL UNIVERSITY

Email address: hlee014@snu.ac.kr

KAZUSHI UEDA, GRADUATE SCHOOL OF MATHEMATICAL SCIENCES, THE UNIVERSITY OF TOKYO

Email address: kazushi@ms.u-tokyo.ac.jp